

Supplementary Material for “An ADMM-based Battery Dispatch Optimization Approach for Electric Vehicle Aggregators with Input-Discriminative Local Differential Privacy Budgets”

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I. DIFFERENTIAL PRIVACY-RELATED DEFINITIONS

Differential privacy is a mathematical framework for ensuring the privacy of individuals in datasets [40]. It can provide a strong guarantee of privacy by allowing data to be analyzed without revealing sensitive information about any individual in the dataset. For a V2G-based BDO problem, the concerned query mechanism for a sensitive vector $\mathbf{d}$ needs to return similar $\mathbf{d}$ data on adjacent datasets $\mathbf{d}$ and $\mathbf{d}'$ for each EV in a probabilistic manner. The formal definition is given below:

Definition 1 (*Differential Privacy*[40]): Given a value $\rho \in \mathbb{R}_+^n$, a randomized V2G-based BDO mechanism $\mathcal{M} : \mathcal{D} \rightarrow \mathcal{R}$ with domain $\mathcal{D}$ and range $\mathcal{R}$ is ε -differential private if, for any output $\mathcal{O}$ and any two adjacent inputs $\mathbf{d} \sim_\rho \mathbf{d}' \in \mathbb{R}^n$ satisfy

$$\mathbb{P}[\mathcal{M}(\mathbf{d}) = \mathcal{O}] \leq e^\varepsilon \mathbb{P}[\mathcal{M}(\mathbf{d}') = \mathcal{O}], \quad (\text{I-1})$$

where $\mathbb{P}$ denotes the probability over runs of $\mathcal{M}$, and range $\mathcal{R}$ denotes the feasibility space $\mathbf{X}$. Adjacent inputs $\mathbf{d} \sim_\rho \mathbf{d}'$ is equivalent to $\exists i, s.t. |\mathbf{d}_i - \mathbf{d}'_i| \leq \rho$ and $\mathbf{d}_j = \mathbf{d}'_j, \forall j \neq i$. The domain $\mathcal{D}$ includes all feasible datasets of $\mathbf{d}$. And $\mathbf{X}$ indicates the feasibility space of the conventional V2G-based BDO problem in (2). The parameter ε is used to control the level of privacy. In other words, there are smaller values of ε , thus providing stronger privacy protection.

Definition 2 (ℓ_1 -sensitivity[40]): The ℓ_1 -sensitivity of the V2G-based BDO mechanism $\mathcal{M}$ on ρ -adjacent datasets, denoted by Δ_ρ with the 1-norm calculation, is defined as:

$$\Delta_\rho = \max_{\mathbf{d} \sim_\rho \mathbf{d}'} \|\mathcal{M}(\mathbf{d}) - \mathcal{M}(\mathbf{d}')\|_1. \quad (\text{I-2})$$

With available Δ_ρ , the Gaussian mechanism is expressed as below:

Definition 3 (*Gaussian mechanism* [40]): Let the V2G-based BDO mechanism $\mathcal{M}$ be an algorithm with ℓ_1 -sensitivity Δ_ρ that maps the datasets $\mathbf{d}$ to $\mathbb{R}^n$. Let $\varepsilon \in (0, 1)$ and $\gamma \geq 2 \ln(1.25/\varepsilon)$, the noise ξ follows the Normal distribution with 0 mean and standard deviation $\sigma \geq \gamma \Delta_\rho / \varepsilon$. The Gaussian mechanism $\mathcal{M}(\mathbf{d}) = \mathcal{M}(\mathbf{d}) + \xi$ attains ε -differential privacy.

This definition of DP is suitable for cases where one uniform level of privacy needs to across all elements in the databases. Consider the V2G-based BDO mechanism $\mathcal{M}$ as a decentralized algorithm $\mathcal{M}(\mathbf{d}_1, \mathbf{d}_2, \dots, \mathbf{d}_{N_E})$, where $\mathbf{d}_i = \mathbf{p}_{sig,i}^t$ for EV $_i$. The decentralized mechanism $\mathcal{M}$ contains N_E local mechanisms $\mathcal{M}_1(\mathbf{d}_1, \mathbf{d}_2, \dots, \mathbf{d}_{N_E}), \dots, \mathcal{M}_{N_E}(\mathbf{d}_1, \mathbf{d}_2, \dots, \mathbf{d}_{N_E})$, where each $\mathcal{M}_i$ represents an individual V2G-based BDO mechanism for EV $_i$. Since $\mathcal{M}_i$ does not necessarily have direct access to the database $\mathbf{d}_j, j \neq i$, $\mathbf{d}_1, \mathbf{d}_2, \dots, \mathbf{d}_{N_E}$ can be explicitly expressed as the input to all local mechanisms. Assume each EV $_i$ can specify the own level of privacy ε_i , and we formalize this specification in the LDP definition:

Definition 4 (*Local Differential Privacy*[22]): With privacy budget ε_i for EV $_i$, $\mathcal{M}_i$ provides ε_i -LDP if, for any output $\mathcal{O}_i$ under any two adjacent inputs $\mathbf{d}_i \sim_\rho \mathbf{d}'_i \in \mathbb{R}$ satisfy

$$\mathbb{P}[\mathcal{M}_i = \mathcal{O}_i | \mathbf{d}_i] \leq e^{\varepsilon_i} \mathbb{P}[\mathcal{M}_i = \mathcal{O}_i | \mathbf{d}'_i], \quad (\text{I-3})$$

where adjacent inputs $\mathbf{d}_i \sim_\rho \mathbf{d}'_i$ is equivalent to $|\mathbf{d}_i - \mathbf{d}'_i| \leq \rho$.

For EV $_i$, LDP regards $\mathbf{d}_i$ during all time periods as equally sensitive, which is not practical for EV customers' privacy protection in different time periods. For example, the availability of from V2G-based BDO dispatch support is very essential for stable grid operations during peak load periods, but this availability may be not sensitive during low load demand periods. To feature this sensitivity difference, ID-LDP is the variant of LDP for which a privacy budget

ε_i can be a function with respect to time period t . To address this issue, the definition of ID-LDP is proposed as follows:

Definition 5 (*Input-discriminative Local Differential Privacy*[24]): Let ε_i^t be the specific privacy budget for EV $_i$ in the time period t , and define $r(\varepsilon_i^t)$ be a function of ε_i^t in time period t . Then, $\mathcal{M}_i$ provides $r(\varepsilon_i^t)$ -ID-LDP if, for any output $\mathcal{O}_i^t$ under any two adjacent inputs $\mathbf{d}_i^t \sim_\rho \mathbf{d}'_i^t \in \mathbb{R}$ satisfy

$$\mathbb{P}[\mathcal{M}_i = \mathcal{O}_i^t | \mathbf{d}_i^t] \leq e^{r(\varepsilon_i^t)} \mathbb{P}[\mathcal{M}_i = \mathcal{O}_i^t | \mathbf{d}'_i^t], \quad (\text{I-4})$$

where the privacy budget function $r(\varepsilon_i^t)$ can vary the obfuscation levels of EV $_i$ in different time periods.

II. PROOF OF THEOREM 1

There exists an optimal BDO solution $\mathbf{x}_i^{t*}$ with the minimum objective function value $\mathbf{F}^*$ for the C-BDO model (1a)-(1d). We incorporate $\mathbf{p}_{sig,i}^{t*} \in \mathbb{R}$ such that $\sum_{i=1}^{N_E} \mathbf{p}_{sig,i}^{t*} = \mathbf{p}_{ref}^t$. Reformulate $\|\cdot\|_2$ in the objective function (1a) by

$$\left\| \sum_{i=1}^{N_E} \mathbf{c}^T \mathbf{x}_i^{t*} - \mathbf{p}_{ref}^t \right\|_2 = \left\| \sum_{i=1}^{N_E} (\mathbf{c}^T \mathbf{x}_i^{t*} - \mathbf{p}_{sig,i}^{t*}) \right\|_2 \quad (\text{II-5})$$

Then, we achieve optimality by

$$\sum_{t=1}^{N_T} \left\| \sum_{i=1}^{N_E} (\mathbf{c}^T \mathbf{x}_i^{t*} - \mathbf{p}_{sig,i}^{t*}) \right\|_2 = \sum_{t=1}^{N_T} \left\| \sum_{i=1}^{N_E} \mathbf{c}^T \mathbf{x}_i^{t*} - \mathbf{p}_{ref}^t \right\|_2 = \mathbf{F}^* \quad (\text{II-6})$$

This proves that $(\mathbf{x}_i^{t*}, \mathbf{p}_{sig,i}^{t*}) \in \mathbf{X}$ can be the optimal BDO solution with the minimum objective function value $\mathbf{F}^*$.

Due to the triangle inequality of norms, i.e., $|a + b| \leq |a| + |b|$ for any real-valued numbers a and b , we obtain that $\mathbf{F}^* \leq \mathbf{F} \leq \mathbf{G}$ naturally holds. This suggests that $\mathbf{F}^*$ can be the lower bound of $\mathbf{G}$ over $\mathbf{X}$. In other words, $(\mathbf{x}_i^{t*}, \mathbf{p}_{sig,i}^{t*}) \in \mathbf{X}$ can be the optimal solution iff the minimum function value $\mathbf{G}^* = \mathbf{F}^*$ is achieved, which demonstrates $(\mathbf{x}_i^{t*}, \mathbf{p}_{sig,i}^{t*})$ can also be the optimal BDO solution for minimizing the objective function $\mathbf{G}$ over the feasibility space $\mathbf{X}$. This proves Theorem 1. ■

III. PROOF OF THEOREM 2

The objective is to prove that the optimal charge-discharge solution $\mathbf{x}_i^{t\dagger} = \mathbf{x}_i^{t*}$ holds for EV $_i$. It is clear that the following (III-7) stands.

$$\mathbf{s}_i^{t*} = \mathbf{c}^T \mathbf{x}_i^{t*} - \mathbf{p}_{sig,i}^{t*} + \alpha_i^{t\dagger} \xi_i^t - \alpha_i^{t\dagger} \xi_i^t. \quad (\text{III-7})$$

We denote $\tilde{\mathbf{p}}_{sig,i}^{t\dagger} = \mathbf{p}_{sig,i}^{t*} + \alpha_i^{t\dagger} \xi_i^t$. Then, we have

$$\mathbf{s}_i^{t*} = \mathbf{c}^T \mathbf{x}_i^{t*} - \tilde{\mathbf{p}}_{sig,i}^{t\dagger} + \alpha_i^{t\dagger} \xi_i^t \quad (\text{III-8})$$

As $\mathbf{s}_i^{t\dagger} \geq \mathbf{s}_i^{t*}$ and $\mathbf{y}_i^{t\dagger} = \mathbf{s}_i^{t\dagger}$ achieved at optimality, minimizing the objective function $\mathbf{g}_i$ ensures $\mathbf{s}_i^{t\dagger} = \mathbf{s}_i^{t*}$, where other two terms in $\mathbf{g}_i$ can be held at convergence, i.e.,

$$\mu^t (\tilde{\mathbf{p}}_{sig,i}^{t\dagger} - \alpha_i^{t\dagger} \xi_i^t) = \mu^t \mathbf{p}_{sig,i}^{t*} \quad (\text{III-9})$$

and

$$\frac{\phi}{2} \left\| \sum_{j=1, j \neq i}^{N_E} \tilde{\mathbf{p}}_{sig,j}^{t\dagger} + \tilde{\mathbf{p}}_{sig,i}^{t\dagger} - \mathbf{p}_{ref}^t \right\|_2^2 = \frac{\phi}{2} \left\| \sum_{j=1, j \neq i}^{N_E} \mathbf{p}_{sig,j}^{t*} + \mathbf{p}_{sig,i}^{t*} - \mathbf{p}_{ref}^t \right\|_2^2 \approx 0 \quad (\text{III-10})$$

Thus, we rewrite $s_i^{t\dagger} = \mathbf{c}^T \mathbf{x}_i^{t*} - \tilde{\mathbf{p}}_{sig,i}^{t\dagger} + \boldsymbol{\alpha}_i^{t\dagger} \boldsymbol{\xi}_i^t$. This suggests that $\mathbf{x}_i^{t\dagger} = \mathbf{x}_i^{t*}$ holds with obfuscated $\tilde{\mathbf{p}}_{sig,i}^{t\dagger} = \mathbf{p}_{sig,i}^{t*} + \boldsymbol{\alpha}_i^{t\dagger} \boldsymbol{\xi}_i^t$ for the privacy-preserving BDO model. ■

IV. PROOF OF THEOREM 3

We prove that $\mathcal{M}$ is $r_{\varepsilon,i}^t$ -ID-LDP with respect to the local dataset of sub-problem i during all iterations. Let the query output answer for the sub-problem i be $\mathcal{O}_i^t = \tilde{\mathbf{p}}_{sig,i}^t$, and we alternatively rewrite (IV-11) in the definition of $r_{\varepsilon,i}^t$ -LDP for the sub-problem i :

$$\mathbb{P}[\mathcal{M}_i = \mathcal{O}_i^t | \mathbf{x}_i^t] \leq \mathbb{P}[\mathcal{M}_i = \mathcal{O}_i^t | \mathbf{x}_i^{t'}] e^{r_{\varepsilon,i}^t}, \quad (\text{IV-11})$$

for two $r_{\varepsilon,i}^t$ -indistinguishable charge-discharge solutions $\mathbf{x}_i^t$ and $\mathbf{x}_i^{t'}$ in time period t . For convenience, the query output $\mathcal{O}_i^t$ of EV $_i$ for $\forall t \in \mathcal{T}$ with a random noise $\boldsymbol{\xi}_i^t$ can be written as $\mathcal{O}_i^t = \tilde{\mathbf{p}}_{sig,i}^t = \mathbf{p}_{sig,i}^t + \boldsymbol{\alpha}_i^t \boldsymbol{\xi}_i^t$ where $\forall t \in \mathcal{T}$.

Therefore, the ratio of probabilities on two ε_i -indistinguishable charge-discharge solutions $\mathbf{x}_i^t$ and $\mathbf{x}_i^{t'}$ can be bounded by

$$\begin{aligned} \frac{\mathbb{P}[\mathcal{M}_i = \mathcal{O}_i^t | \mathbf{x}_i^t]}{\mathbb{P}[\mathcal{M}_i = \mathcal{O}_i^t | \mathbf{x}_i^{t'}]} &= \frac{\mathbb{P}[\mathbf{p}_{sig,i}^t + \boldsymbol{\alpha}_i^t \boldsymbol{\xi}_i^t = \mathcal{O}_i^t | \mathbf{x}_i^t]}{\mathbb{P}[\mathbf{p}_{sig,i}^{t'} + \boldsymbol{\alpha}_i^t \boldsymbol{\xi}_i^t = \mathcal{O}_i^t | \mathbf{x}_i^{t'}]} \\ &= \frac{\mathbb{P}[\boldsymbol{\alpha}_i^t \boldsymbol{\xi}_i^t = \mathcal{O}_i^t - \mathbf{p}_{sig,i}^t | \mathbf{x}_i^t]}{\mathbb{P}[\boldsymbol{\alpha}_i^t \boldsymbol{\xi}_i^t = \mathcal{O}_i^t - \mathbf{p}_{sig,i}^{t'} | \mathbf{x}_i^{t'}]} \stackrel{(i)}{=} \frac{\boldsymbol{\alpha}_i^t \exp\{\frac{r_{\varepsilon,i}^t \|\mathcal{O}_i^t - \mathbf{p}_{sig,i}^t\|_1}{\Delta_{\rho,i}}\}}{\boldsymbol{\alpha}_i^t \exp\{\frac{r_{\varepsilon,i}^t \|\mathcal{O}_i^t - \mathbf{p}_{sig,i}^{t'}\|_1}{\Delta_{\rho,i}}\}} \\ &= \exp\left(\frac{r_{\varepsilon,i}^t \|\mathcal{O}_i^t - \mathbf{p}_{sig,i}^t\|_1 - r_{\varepsilon,i}^t \|\mathcal{O}_i^t - \mathbf{p}_{sig,i}^{t'}\|_1}{\Delta_{\rho,i}}\right) \\ &\stackrel{(ii)}{\leq} \exp\left(\frac{r_{\varepsilon,i}^t \|\mathbf{p}_{sig,i}^t - \mathbf{p}_{sig,i}^{t'}\|_1}{\Delta_{\rho,i}}\right) \stackrel{(iii)}{\leq} \exp\left(\frac{r_{\varepsilon,i}^t \Delta_{\rho,i}}{\Delta_{\rho,i}}\right) = e^{r_{\varepsilon,i}^t}, \end{aligned}$$

where (i) comes from the definition of the probability density function of the Gaussian distribution. In (ii) step, it is followed by the inequality of norms, i.e., $|a| - |b| \leq |a - b|$ for any real-valued numbers a and b . For the (iii) step, $\|\mathbf{p}_{sig,i}^t - \mathbf{p}_{sig,i}^{t'}\|_1$ denotes the ℓ_1 -sensitivity on ρ -indistinguishable output datasets $\mathbf{p}_{sig,i}^t$ and $\mathbf{p}_{sig,i}^{t'}$ subject to $\|\mathbf{p}_{sig,i}^t - \mathbf{p}_{sig,i}^{t'}\|_1 \leq \Delta_{\rho,i}$.

Accordingly, it is clear that (IV-11) holds based on (IV-12), which proves this Theorem. ■